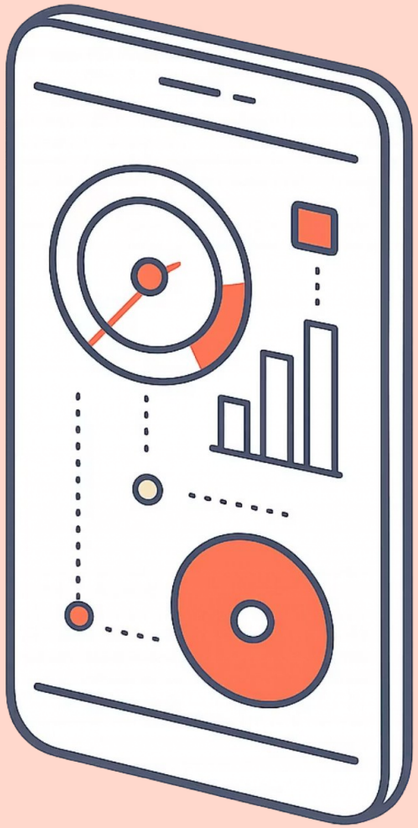

Human Activity Recognition System

From Smartphone Sensor Data — Final Year Predictive Analytics Project

Team: ARDRA SELIN A G | ARJUN S | ARYA SUKU

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PROBLEM STATEMENT

Why Human Activity Recognition Matters

Healthcare

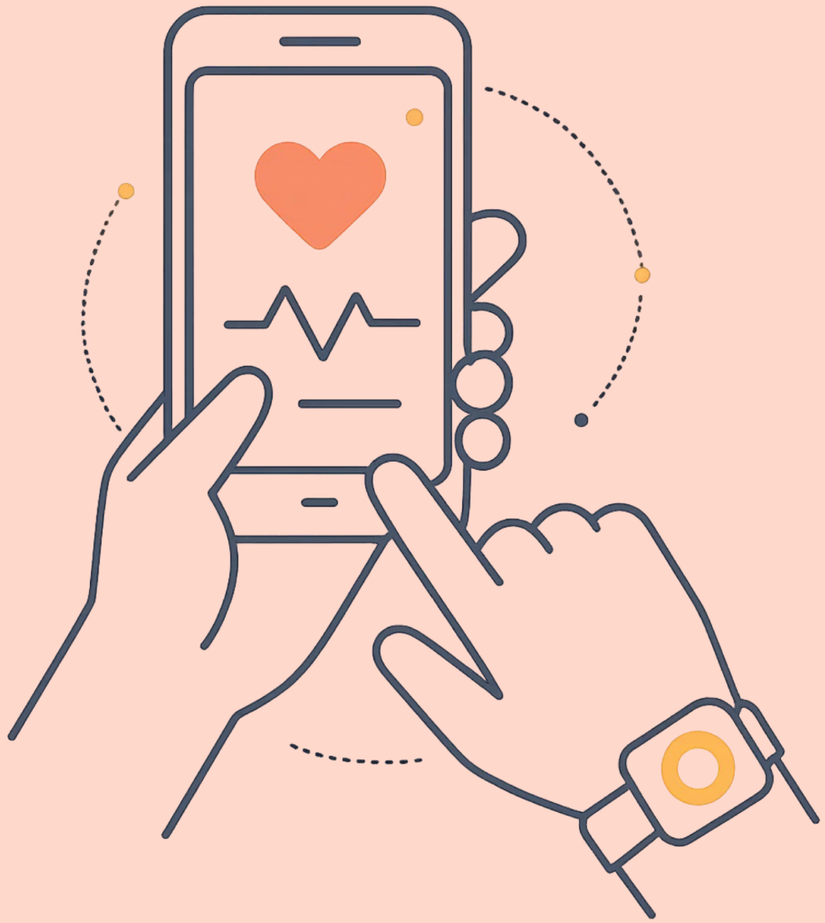
Monitor elderly patients, detect falls, and track rehabilitation progress remotely.

Fitness Tracking

Automatically log workouts, steps, and calories for personalized health insights.

Smart Devices

Enable context-aware smartphones and wearables that adapt to user behavior.



PROJECT OBJECTIVES

What We Set Out to Achieve

Classify Activities

Build a system that accurately identifies human activities from smartphone sensor data.

Compare ML Models

Evaluate multiple machine learning algorithms to determine the best-performing approach.

Deploy a Working System

Create an interactive, real-time prediction interface accessible to end users.

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DATASET OVERVIEW

UCI Human Activity Recognition Dataset

561 Features

Derived from time and frequency domain signals

30 Participants

Volunteers aged 19–48 performing recorded activities

6 Activity Classes

Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, Lying

Accelerometer & Gyroscope

3-axial data collected at 50Hz via Samsung Galaxy S3

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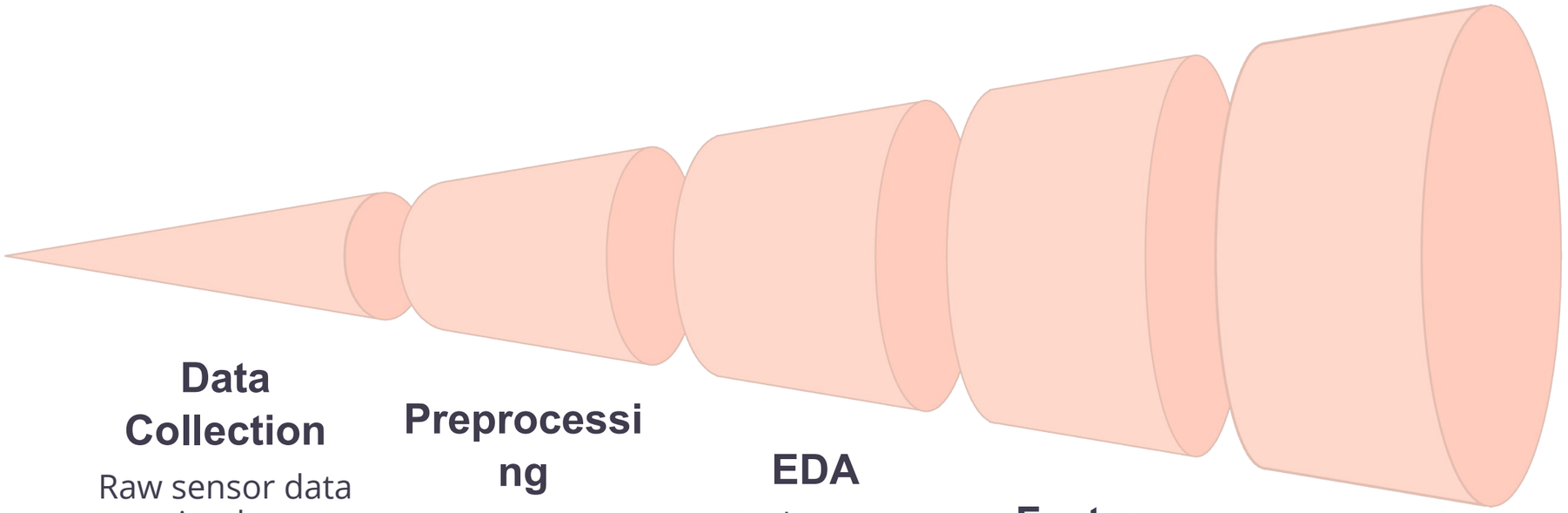
METHODOLOGY

End-to-End Project Pipeline



Data Collection

Raw sensor data intake

Preprocessing

Cleaning and normalization

EDA

Exploratory analysis

Feature Engineering

Create predictive features

Model Training

Train and tune models

The pipeline moves sequentially from raw sensor data collection through rigorous preprocessing and exploration, into model development, and finally to a deployed interactive system.

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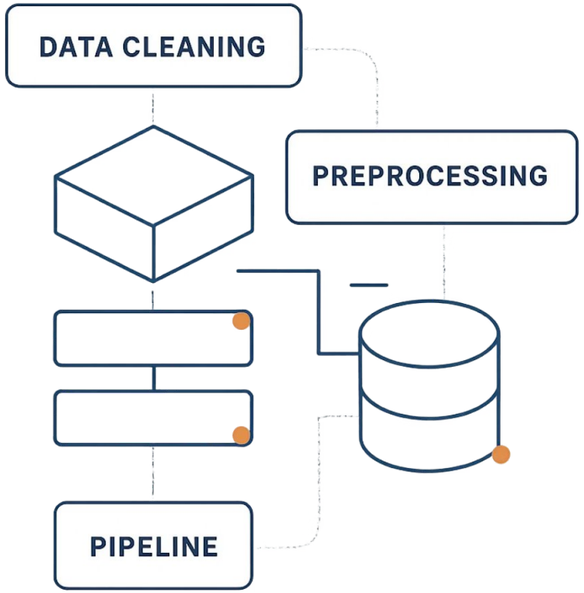
DATA PREPARATION

Preprocessing & Exploratory Data Analysis



Preprocessing Steps

Data Cleaning: Removed nulls, duplicates, and inconsistent entries
Label Encoding: Converted activity labels to numeric classes (0–5)
Feature Scaling: Standardized features to prevent scale bias

EDA Key Insights

- Class distribution was well-balanced across all 6 activities
- Correlation heatmap revealed redundant features for dimensionality reduction
- Accelerometer signals showed distinct patterns for static vs. dynamic activities

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FEATURE ENGINEERING

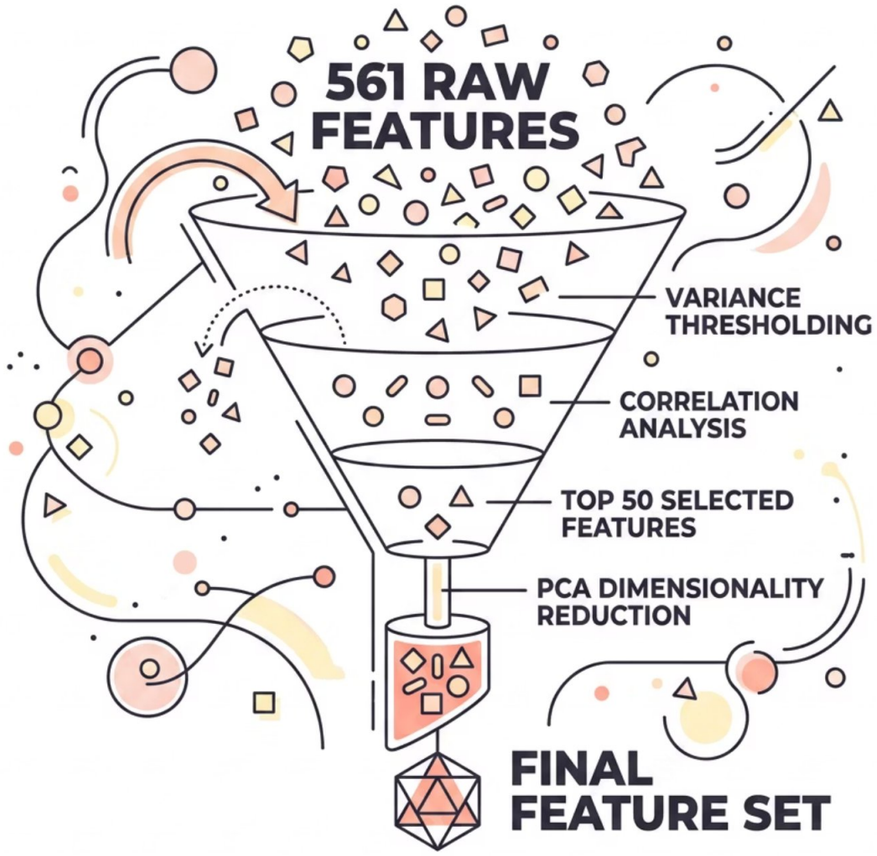
Selecting the Right Signals

Feature Selection

From 561 raw features, the **top 50 most informative** were selected using variance thresholding and correlation analysis to reduce noise and improve model efficiency.

Dimensionality Reduction

Principal Component Analysis (PCA) was explored to visualize class separability and confirm that the selected features captured the majority of variance in the data.



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MODELS & EVALUATION

Four Models, One Best Performer

Logistic Regression

Baseline linear classifier

Decision Tree

Interpretable rule-based model

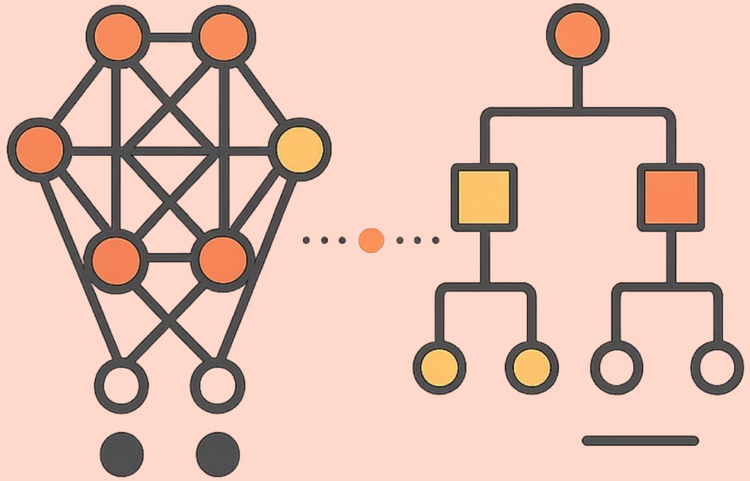
Random Forest ★

Ensemble of decision trees

K-Nearest Neighbors

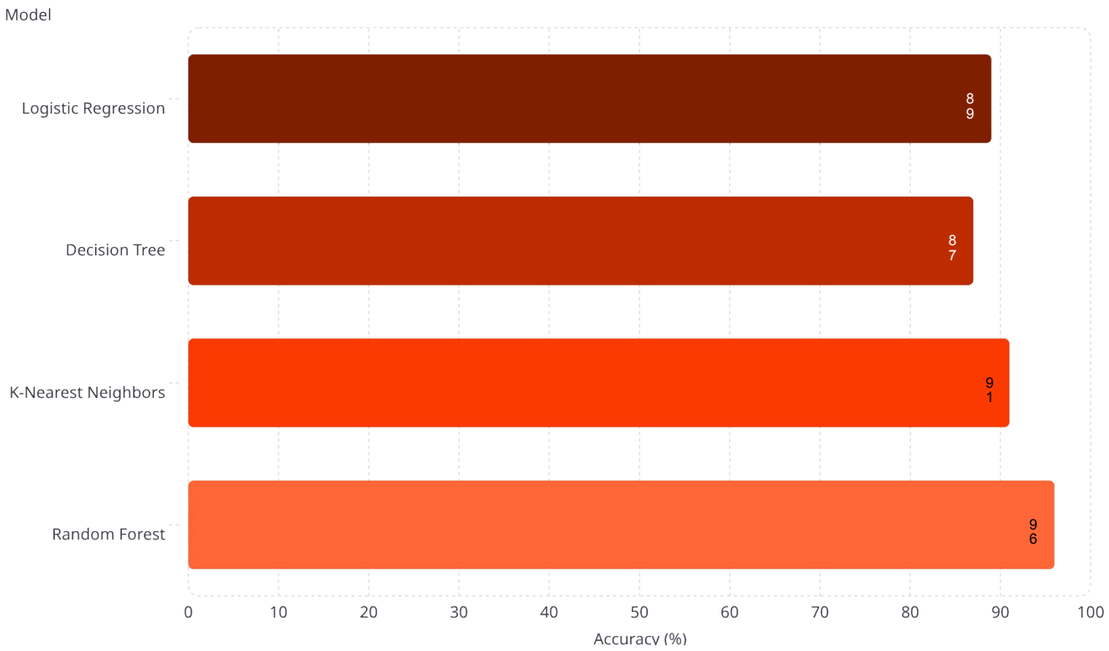
Distance-based classifier

Performance was measured across **Accuracy, Precision, Recall,** and **F1-Score** to ensure a comprehensive evaluation beyond a single metric.



RESULTS

Model Performance Comparison



Why Random Forest Won

- Ensemble approach reduces overfitting by averaging multiple trees
- Handles high-dimensional sensor data robustly
- Built-in feature importance revealed accelerometer signals as most predictive

Best F1-score of **0.96**, confirming strong precision and recall balance

Best Result: Random Forest achieved **~96% accuracy** — the highest among all tested models.

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DEPLOYMENT & CONCLUSION

From Model to Real-World Application

Streamlit Deployment

An interactive web app accepts live sensor input and returns real-time activity predictions with a clean, user-friendly interface.

⚠ Challenges Faced

High dimensionality (561 features) required careful selection; similar activities (e.g., sitting vs. standing) demanded fine-grained feature discrimination.

Future Directions

Explore LSTM deep learning for temporal patterns, integrate real-time mobile sensor streaming, and expand to more complex activity classes.

Thank you for your attention. This project demonstrates that smartphone sensor data, combined with ensemble machine learning, can power accurate, deployable human activity recognition systems.

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